

Specialization: *Health Analytics*

Business Focus: *Health Data Analysis*

Tool: *Microsoft Power BI*

Healthcare Analytics: Optimizing Patient Flow and Revenue at Pure Life Hospital

Project Learning Opportunities

This project gives analysts hands-on experience in the end-to-end healthcare BI lifecycle. Participants will build a relational Star Schema from flat transactional data, use advanced DAX to calculate key clinical and financial KPIs, and design tailored dashboards for both executive-level tracking and granular patient reviews.

Tools and Technology to be Used



Case Study Overview

Introduction to the Business

Pure Life Hospital is a comprehensive healthcare facility offering a wide range of medical services. Based on operational data, the hospital is equipped to provide care across multiple settings, **including Inpatient, Outpatient, and Emergency.**

The hospital operates across 8 specialized departments: **Cardiology, Pediatrics, Surgery, Dermatology, Neurology, Orthopedics, General Practice, and Oncology.** Financially, the hospital relies heavily on a mix of private payments and **Health Maintenance Organizations (HMOs)**, partnering with major providers such as **Avon HMO, Reliance HMO, NHIS, AXA Mansard, and Hygeia HMO.**

The hospital sees a significant volume of patients with varying severity levels (ranging from 1 to 10), managing complex cases that result in diverse outcomes, including **discharges, ongoing admissions, transfers, and mortality.**





Case Study Overview

Problem Statement

Despite processing a high volume of patients and generating substantial billed revenue, **Pure Life Hospital** lacks a centralized, data-driven mechanism to monitor the intersection of clinical outcomes and financial performance. Specifically, the hospital struggles with:

- **Revenue Cycle Visibility:** Tracking the status of payments (Paid, Pending, Rejected) and understanding the exact burden of "Out-of-Pocket" costs versus "Insurance Covered" amounts.
- **Resource Allocation:** Without real-time tracking of Visit Type and Department volume, the hospital cannot accurately predict staffing needs or bed capacity.
- **Clinical Quality Tracking:** There is a need to systematically monitor Severity Scores against Outcome Status (e.g., Deceased vs. Discharged) to evaluate the quality of care across different departments and doctors.
- **Patient History Management:** Clinicians lack a quick, consolidated view of a returning patient's historical diagnoses and visits, leading to fragmented care.

Rationale for the Project

(What is the Importance of the project to the business)

1.

Financial Rationale

By visualizing Payment Status and Total_Billed_Amount, the finance team can aggressively target "Pending" or "Rejected" claims, improving cash flow and reducing revenue leakage.

2.

Operational Rationale

Tracking metrics like Average Length of Stay (derived from Admission Date and Discharge Date) will allow administrators to optimize bed turnover rates and reduce operational bottlenecks.

3.

Clinical Rationale

The Patient Detail dashboard will allow doctors to instantly pull up a Patient_ID, view their Severity Score history, and see previous Primary Diagnosis Descriptions, ensuring faster and more accurate clinical decision-making.



Project Workflow

STEP 1

Data Loading & Cleaning

Load data, clean and structure it for analysis.



STEP 2

Data Modeling

Breaking the flat file into different tables for modeling



STEP 3

M Language & DAX

Generate insight from the data



STEP 4

Visualization / Dashboard

Design an interactive dashboard with relevant insights and KPIs using slicers and timeline.



STEP 5

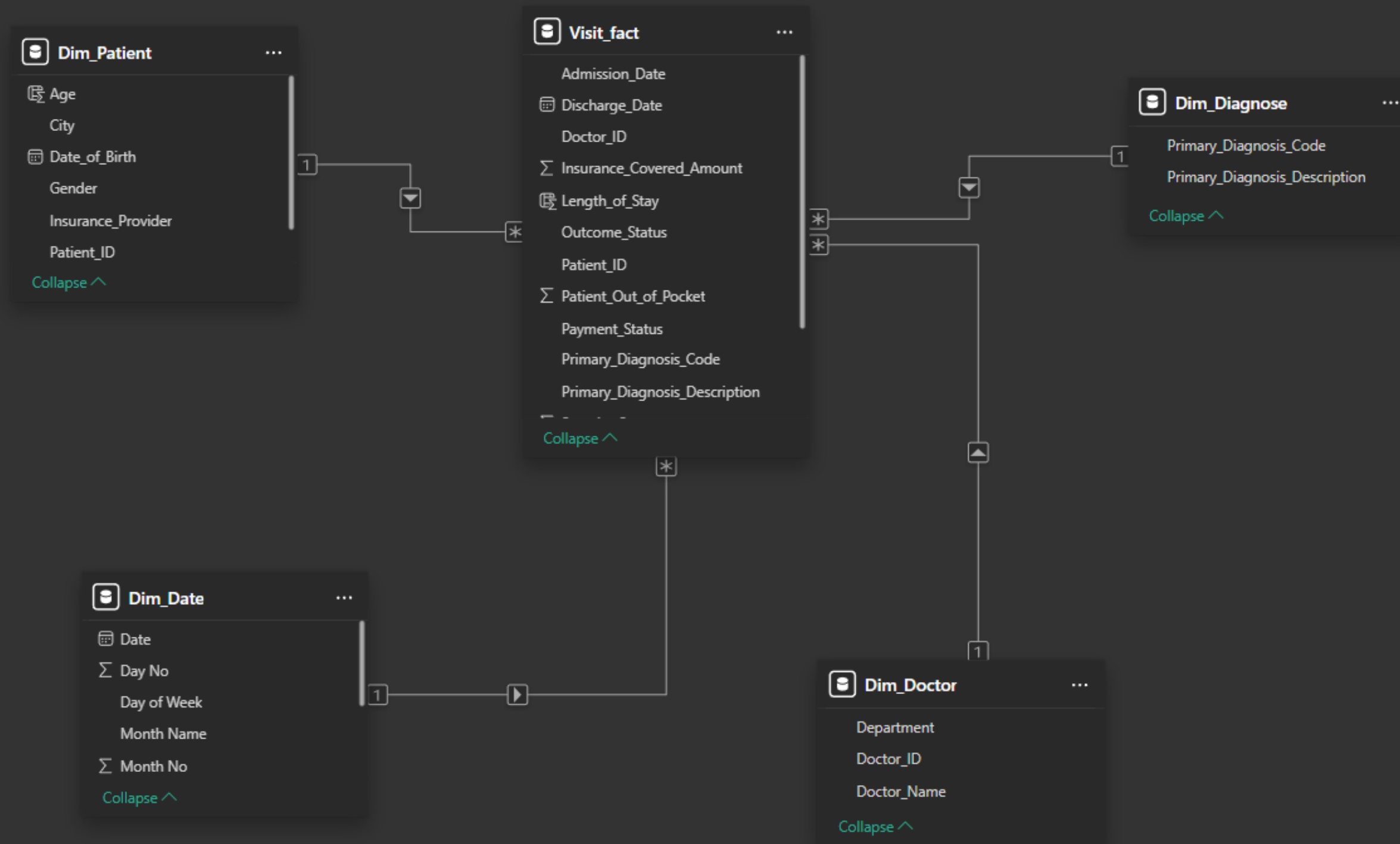
Recommendations

Actionable strategies the business can implement in order to experience growth.

Data Description

- **Patient_ID:** A unique alphanumeric identifier assigned to each patient.
- **Patient_Name:** The full legal name of the patient.
- **Date_of_Birth:** The patient's birth date, used to calculate age and demographic trends.
- **Gender:** The registered gender of the patient (Male/Female).
- **City:** The patient's city of residence.
- **Visit_ID:** A unique identifier for every single hospital visit/encounter.
- **Visit_Type:** The categorization of the visit (Inpatient, Outpatient, or Emergency).
- **Admission_Date / Discharge_Date:** The dates the patient entered and exited the hospital's care. Used to calculate the length of Stay.
- **Department:** The specific medical unit where the patient received care (e.g., Cardiology, Surgery).
- **Patient_Out_of_Pocket:** The remaining financial balance that the patient is personally responsible for paying.
- **Payment_Status:** The current standing of the bill (Paid, Pending, or Rejected).
- **Doctor_ID / Doctor_Name:** The unique ID and name of the primary attending physician.
- **Primary_Diagnosis_Code / Description:** The standardized code (e.g., K53) and text description of the patient's primary ailment.
- **Severity_Score:** A numerical rating (1-10) indicating the critical nature of the patient's condition.
- **Outcome_Status:** The final result of the visit (Discharged, Admitted, Transferred, or Deceased).
- **Total_Billed_Amount:** The gross financial cost of the hospital visit before insurance deductions.
- **Insurance_Provider:** The HMO or private insurance company covering the patient (Note: Contains some missing/null values indicating self-pay).
- **Insurance_Covered_Amount:** The specific portion of the bill approved and paid by the Insurance Provider.

Model: Star Schemas



DASHBOARD 1: General Overview



DASHBOARD 2: Patient Information



Pages

Overview

Patients

Report

Patient Information Dashboard

Search



Patient_ID	Patient_Name	Gender	Age	City	Visit_Type	Insurance_Provider	Billed Amount	Insurance Covered Amount	Severity_Score	Admission_Date	Discharge
P-1014	Roger Rocha	Female	47	Richardsonland	Outpatient	NHIS	\$41,552	\$33,241.6	2	Sunday, January 01, 2023	Sunday, Jan
P-1326	Christine Nunez	Male	64	West Elizabeth	Outpatient	None	\$11,896	\$0	10	Sunday, January 01, 2023	Sunday, Jan
P-1459	Juan Garrett	Male	42	Port Joel	Outpatient	Private	\$36,992	\$29,593.6	7	Sunday, January 01, 2023	Sunday, Jan
P-1660	Kevin Howell	Male	21	Careyville	Outpatient	Avon HMO	\$19,610	\$15,688	9	Sunday, January 01, 2023	Sunday, Jan
P-2271	Donald Malone	Female	62	East Danafurt	Outpatient	NHIS	\$33,816	\$27,052.8	6	Sunday, January 01, 2023	Sunday, Jan
P-1062	Lisa Roberts	Male	63	Maxwellburgh	Outpatient	Hygeia HMO	\$9,208	\$7,366.4	1	Monday, January 02, 2023	Monday, Jan
P-1162	Kenneth Austin	Male	56	East Jessicaburgh	Emergency	NHIS	\$82,324	\$65,859.2	7	Sunday, January 01, 2023	Monday, Jan
P-1182	Christian Perez	Male	47	West Jennifer	Outpatient	Avon HMO	\$46,862	\$37,489.6	10	Monday, January 02, 2023	Monday, Jan
P-1266	Samantha Davis	Female	21	Jerryton	Outpatient	Avon HMO	\$39,378	\$31,502.4	3	Monday, January 02, 2023	Monday, Jan
P-1380	Zachary Jackson	Male	7	East Allisonville	Outpatient	Reliance HMO	\$7,969	\$6,375.2	3	Monday, January 02, 2023	Monday, Jan
P-1425	Amanda Jimenez	Male	85	Lake Stephanieport	Outpatient	NHIS	\$42,233	\$33,786.4	3	Monday, January 02, 2023	Monday, Jan
P-1737	Carolyn Galloway	Female	74	Bowmanshire	Outpatient	NHIS	\$22,652	\$18,121.6	1	Monday, January 02, 2023	Monday, Jan
P-1801	Garrett Griffin	Female	27	North James	Outpatient	Private	\$26,752	\$21,401.6	2	Monday, January 02, 2023	Monday, Jan
P-2052	Scott Cox	Male	69	Potterport	Emergency	Private	\$45,410	\$36,328	5	Monday, January 02, 2023	Monday, Jan
P-2101	William Becker	Male	69	Jonesfurt	Outpatient	None	\$37,944	\$0	10	Monday, January 02, 2023	Monday, Jan
P-2428	Carol Taylor	Male	23	West Julie	Outpatient	Hygeia HMO	\$49,413	\$39,530.4	3	Monday, January 02, 2023	Monday, Jan
P-1092	Raymond Reynolds	Male	41	Paulland	Outpatient	Reliance HMO	\$21,449	\$17,159.2	2	Tuesday, January 03, 2023	Tuesday, Jan
P-1304	Nicole Atkinson	Female	26	New Colleenview	Inpatient	AXA Mansard	\$87,172	\$69,737.6	1	Tuesday, January 03, 2023	Tuesday, Jan
P-1305	Mr. Michael Davis	Female	4	Hardinghaven	Outpatient	Hygeia HMO	\$14,636	\$11,708.8	6	Tuesday, January 03, 2023	Tuesday, Jan
P-1462	Lee Reilly	Female	16	Joshuaville	Emergency	Avon HMO	\$34,030	\$27,224	10	Monday, January 02, 2023	Tuesday, Jan
P-1603	Sarah Roman	Female	71	Jenkinsmouth	Outpatient	AXA Mansard	\$45,836	\$36,668.8	6	Tuesday, January 03, 2023	Tuesday, Jan

DATASET

Click [Here](#) to download the dataset

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DELVE IN?**

